NICOLA BARILETTO

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GENERAL INFORMATION

PhD candidate in the Department of Statistics and Data Sciences (SDS) at UT Austin.
Academic advisors: Stephen G. Walker (main advisor), Nhat Ho, Alessandro Rinaldo (co-advisors).

EDUCATION

University of Texas at Austin Austin, USA — Aug 2023 - Present
PhD in Statistics.

Bocconi University Milan, Italy — Sep 2020 - Jul 2022
MSc in Economic and Social Sciences (cum laude).

Princeton University Princeton, USA — Sep 2019 - Jan 2020
Exchange semester at the Department of Politics.

Bocconi University Milan, Italy — Sep 2017 - Jul 2020
BSc in International Politics and Government (cum laude).

PRE-DOCTORAL RESEARCH EXPERIENCE

Bocconi University Milan, Italy — Sep 2022 - Aug 2023
Pre-doctoral researcher at ERC LOSS project (PI: Catherine E. De Vries).

Bocconi University Milan, Italy — May 2022 - Jul 2022
Research assistant for Pierpaolo Battigalli.

Princeton University Princeton, USA — Oct 2019 - Aug 2021
Research assistant for Jacob N. Shapiro.

TEACHING EXPERIENCE

DSC 381 – Probability and Simulation-Based Inference — Aug 2025 - Present
Co-developer of, and head teaching assistant for MSc course led by Jay Bartroff (instructor) at UT Austin.

HONORS & AWARDS

Summer 2026 Continuing Fellowship
UT Austin Graduate School merit-based fellowship covering tuition, health insurance and stipend.

2024/2025 Graduate Dean's Prestigious Supplement Fellowship
UT Austin Graduate School supplement in recognition of prestigious "Giorgio Mortara" PhD Scholarship.

Best poster award
14th International Conference on Bayesian Nonparametrics (UCLA, June 2025)

2023/2024 & 2024/2025 "Giorgio Mortara" Scholarship
Full tuition and stipend PhD scholarship funded by the Bank of Italy.

2020/2021 Bocconi Graduate Merit Award
Full tuition scholarship for 2-year MSc program.

LANGUAGE SKILLS

Italian (native), English (proficient, TOEFL iBT score: 114), Spanish (proficient).

SOFTWARE SKILLS

Python, R, STATA.

PROFESSIONAL SERVICE

I act as a referee for the *Electronic Journal of Statistics, International Conference on Machine Learning, International Conference on Learning Representations, Transactions on Machine Learning Research.*

TALKS

- **BayesComp 2027**, May 2027 (upcoming), College Station (USA) – invited talk.
- **Seminar at LUISS University's Department of AI, Data and Decision Sciences**, December 2026 (upcoming), Rome (Italy) – invited talk.
- **UT Austin's Dept. of Mathematics Junior Probability Seminar**, October 2026, Austin (USA) – board talk.
- **18th ISBA World Meeting**, June–July 2026, Nagoya (Japan) – poster presentation.
- **10th Bayesian Young Statisticians Meeting**, June 2026, Chiba (Japan) – poster presentation.
- **International Symposium on Nonparametric Statistics**, June 2026, Thessaloniki (Greece) – invited talk.
- **ICERM Workshop on Nonparametric Bayesian Inference - Computational Issues**, January 2026, Providence (USA) – lightning talk.
- **14th International Conference on Bayesian Nonparametrics**, June 2025, Los Angeles (USA) – poster presentation.
- **38th Annual Conference on Neural Information Processing Systems**, December 2024, Vancouver (Canada) – poster presentation.
- **41st International Conference on Machine Learning**, July 2024, Vienna (Austria) – poster presentation at *Humans, Algorithmic Decision-Making and Society: Modeling Interactions and Impact* workshop.

ARTICLES UNDER PEER REVIEW

- D. Le, N. Bariletto, A. Rinaldo, and N. Ho (2026). "Partial Differential Equation Barriers to Identifiability in Infinite Mixture Models." *arXiv preprint arXiv:2608.08597*.
- N. Bariletto and S. G. Walker (2026). "On the Geometry of Weak Convergence Without Total Variation Convergence." *arXiv preprint arXiv:2608.03290*.
- N. Bariletto, D. Le, A. Rinaldo, and N. Ho (2026). "Convergence Rates for Latent Mixing Measures in Infinite Homoscedastic Location-Scale Mixture Models." *arXiv preprint arXiv:2605.06845*.
- N. Bariletto and S. G. Walker (2026). "Scalable Posterior Uncertainty for Flexible Density-Based Clustering." *arXiv preprint arXiv:2603.03188*.
- N. Bariletto, H. Nguyen, N. Ho, and A. Rinaldo (2026). "On Bayesian Softmax-Gated Mixture-of-Experts Models." *arXiv preprint arXiv:2604.20551*.
- N. Bariletto and S. G. Walker (2025). "On a Necessary Condition for Posterior Inconsistency: New Insights from a Classic Counterexample." *arXiv preprint arXiv:2510.18126*. – **Minor revision at the Journal of Nonparametric Statistics**.
- N. Bariletto, B. Flores, and S. G. Walker (2025). "Posterior Consistency in Parametric Models via a Tighter Notion of Identifiability." *arXiv preprint arXiv:2504.11360*. – **Invited resubmission at Statistical Science**.

BOOK CHAPTERS

- Agostinelli, C., Aitchison, L., Aldea, E., Allmendinger, R., Ament, S., Anson, B., Arbel, J., Bakshy, E., Balandat, M., Bariletto, N., ... Zhou, Y. (2026). "Handbook of Bayesian Deep Learning." *Zenodo*.

PEER-REVIEWED PUBLICATIONS

- N. Bariletto, N. Ho, and A. Rinaldo (forthcoming). "Conformalized Bayesian Inference, with Applications to Random Partition Models." *Bayesian Analysis*.
- Cremaschi, S., Bariletto, N., and De Vries, C. E. (2025). Without Roots: The Political Consequences of Collective Economic Shocks. *American Political Science Review*, 119(4):1963–1982
- Bariletto, N. and Ho, N. (2024). Bayesian Nonparametrics Meets Data-Driven Distributionally Robust Optimization. *Advances in Neural Information Processing Systems (NeurIPS 2024)*, 38
- Nguyen, K., Bariletto, N., and Ho, N. (2024). Quasi-Monte Carlo for 3D Sliced Wasserstein. *International Conference on Learning Representations (ICLR 2024, Spotlight)*, 12
- Siwakoti, S., Yadav, K., Bariletto, N., Zanotti, L., Erdogdu, U., and Shapiro, J. N. (2021a). How COVID drove the evolution of fact-checking. *Harvard Kennedy School Misinformation Review*

OTHER PUBLICATIONS

- Siwakoti, S., Yadav, K., Thange, I., Bariletto, N., Zanotti, L., Ghoneim, A., and Shapiro, J. (2021b). Localized misinformation in a global pandemic: Report on COVID-19 narratives around the world. *Empirical Studies of Conflict Project, Princeton, NJ: Princeton University*
- Thange, I., Bariletto, N., Zanotti, L., Rob, J., Siwakoti, S., and Shapiro, J. N. (2020). How Russia, China, and other governments use coronavirus disinformation to reshape geopolitics. *The Bulletin of the Atomic Scientists*